

Problem Set 2 – AI Weather-Bot 3000

Kshitiz Reddy
M.Tech AIML, School of AI, Bennett University

Task 1: Boolean Experiments

```
1 >>> 42 == 42
2 True
3 >>> 'AI' == 'ai'
4 False
5 >>> 10 > 5 and 2 < 1
6 False
```

All three predictions matched actual output. `==` compares equality and is case-sensitive for strings, so `'AI'` and `'ai'` are unequal. In the `and` expression, Python evaluates left to right; `10 > 5` is `True` but `2 < 1` is `False`, so the whole expression is `False`.

Source Code

```
1 """
2 Program: Weather-Bot 3000
3 Purpose: Gives simple lifestyle advice based on temperature and rain.
4 Author: Kshitiz
5 """
6
7 print("Welcome to your AI Climate Assistant.")
8 temp_input = input("What is the current temperature in Celsius? ")
9
10 if temp_input.strip().isdigit():
11     temperature = int(temp_input)
12
13     if temperature > 30:
14         print("AI Alert: It's hot! AI suggests turning on the AC.")
15     elif temperature < 15:
16         print("AI Alert: Chilly! AI suggests a jacket.")
17     else:
18         print("AI Analysis: Temperature is optimal. Enjoy your day!")
19
20     rain_check = input("Is it raining outside? (yes/no) ")
21     if "yes" in rain_check.lower() and temperature < 15:
22         print("AI Recommendation: Stay indoors today, and carry an umbrella.")
23     elif "yes" in rain_check.lower():
24         print("AI Recommendation: Carry an umbrella.")
25 else:
26     print("Input Error: Please enter a valid whole number for temperature.")
```

Console Log

Three runs covering the hot, chilly, and optimal branches.

```
1 Welcome to your AI Climate Assistant.
2 What is the current temperature in Celsius? 35
3 AI Alert: It's hot! AI suggests turning on the AC.
4 Is it raining outside? (yes/no) no
5
```

```
6 Welcome to your AI Climate Assistant.
7 What is the current temperature in Celsius? 10
8 AI Alert: Chilly! AI suggests a jacket.
9 Is it raining outside? (yes/no) yes
10 AI Recommendation: Stay indoors today, and carry an umbrella.
11
12 Welcome to your AI Climate Assistant.
13 What is the current temperature in Celsius? 22
14 AI Analysis: Temperature is optimal. Enjoy your day!
15 Is it raining outside? (yes/no) no
```